

# Studying the Putnam

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Problem Solving Seminar

Department of Mathematics  
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What is the Putnam Exam?

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- afternoon session: 1PM-4PM (6 more problems)



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- not just for supergeniuses or the criminally insane

## Getting Putnam practice



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## Putnam 2016 A1:

Find the smallest integer  $j$  such that for every polynomial  $p(x)$  with integer coefficients

$p^{(j)}(k)$  is divisible by 2016 for all integers  $k$ .

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- try to explore *specific cases*

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Also, polynomials are linear combinations of **monomials**:  $x^n$



$$f(x) = x$$

$$f'(k) = 1$$

$$f''(k) = 0$$

$$f'''(k) = 0$$

$$f''''(k) = 0$$

$$f(x) = x^2$$

$$f'(k) = 2k$$

$$f''(k) = 2$$

$$f'''(k) = 0$$

$$f''''(k) = 0$$

$$f(x) = x^3$$

$$f'(k) = 3k^2$$

$$f''(k) = 6k$$

$$f'''(k) = 6$$

$$f''''(k) = 0$$

$$f(x) = x^4$$

$$f'(k) = 4k^3$$

$$f''(k) = 12k^2$$

$$f'''(k) = 24k$$

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$$f(x) = x^{29}$$

$$f'(k) = 29k^{28}$$

$$f''(k) = 29 \cdot 28k^{27}$$

$$f'''(k) = 29 \cdot 28 \cdot 27k^{26}$$

$$f''''(k) = 29 \cdot 28 \cdot 27 \cdot 26k^{25}$$

How long until divisible by 2016?

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Go one more to get a 2:

$$f^{(6)}(k) = 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25 \cdot 24k^{23}$$

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- Double-check you answered the original question!

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Now, for  $f(x) = x^{55}$  we can calculate

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which is not divisible by 2016, since it isn't divisible by  $2^5$ .

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To prove that 8 is enough, we'll prove a Lemma.

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**Lemma:** If  $n \geq 0$  is an integer and  $f(x) = x^n$ , then  $f^{(j)}(k)$  is divisible by 2016, for any integers  $j$  and  $k$  with  $j \geq 8$ .

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$N = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)$ . Consider the numbers  $\{n, n-1, n-2, n-3, n-4, n-5, n-6, n-7\}$ . At least four of the above numbers must be even, and therefore divisible by 2. Actually, one must be a multiple of 4! Likewise, at least 2 will be multiples of 3. Finally, one of the numbers will be a multiple of 7. Hence  $N$  and  $f^{(j)}(k)$  is divisible by 2016.

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To finish our solution, note that any polynomial will be of the form

$$f(x) = a_0 + a_1x + a_2x^2 + a_nx^n$$

for some integers  $a_0, \dots, a_n$ .

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By our Lemma, when we take the  $j$ 'th derivative for  $j \geq 8$ , each of the powers of  $x$  becomes a multiple of 2016.



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By our Lemma, when we take the  $j$ 'th derivative for  $j \geq 8$ , each of the powers of  $x$  becomes a multiple of 2016.

This makes  $f^{(j)}(k)$  a linear combination of multiples of 2016, and therefore also a multiple of 2016.

## Try it yourself!:

Find the smallest integer  $j$  such that for every polynomial  $p(x)$  with integer coefficients

$p^{(j)}(k)$  is divisible by 2024 for all integers  $k$ .